

CHAPTER 1: SIMPLE HARMONIC MOTION

At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES
1.1	Simple harmonic motion (SHM) a) Define SHM as $a = -\omega^2 y$ b) Sketch the acceleration – displacement ($a - y$) graph. c) Interpret $a - y$ graphs for SHM. d) Define amplitude, A, period T and frequency, f of SHM. e) Sketch graph of displacement – time ($y = A \sin \omega t$) f) Identify amplitude A and period T from graph of displacement – time. g) Use velocity, $v = A\omega \cos \omega t$ and acceleration, $a = -A\omega^2 \sin \omega t$ h) Solve problems related to SHM.

1.1 Simple Harmonic Motion

Periodic Motion is a motion of an object that regularly returns to a given position after a fixed time interval. With a little thought, we can identify several types of periodic motion in everyday life. You go to class every morning. You return to your room each night to sleep. The Moon returns to the same position in its orbit around the Earth each month, a swing in motion and a vibrating tuning fork. In simple words, **periodic motion is a repetitive motion over a fixed period of time.**

Simple harmonic motion (SHM) is a **periodic motion without loss of energy** in which the **acceleration a of a body is directly proportional to its displacement y** from the equilibrium position and is **directed toward equilibrium position but in opposite direction with displacement.**

Explanation	Keywords
Simple harmonic motion is defined as the periodic motion without loss of energy,	Periodic motion No energy loss
in which the acceleration of a body is directly proportional to its displacement from the equilibrium position,	$a \propto y$
and is directed toward equilibrium position but in opposite direction with displacement.	Acceleration a is in opposite direction with displacement y

Equation

$$a = \frac{d^2y}{dt^2} = -\omega^2 y$$

a : Acceleration of the body

ω : Angular frequency

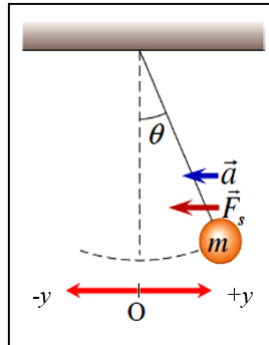
y : Displacement from the equilibrium position

* Negative sign indicates the direction of a is always opposite to the direction of y .

* The angular frequency, ω is always constant thus $a \propto -y$

Examples of Simple Harmonic Motion

1) Simple Pendulum oscillation

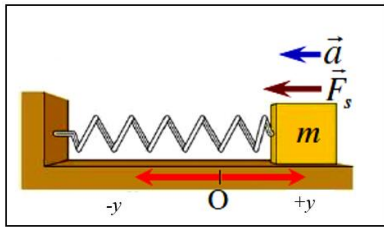


During the oscillation, are there any changes to the magnitude and direction of the acceleration? Explain.

Simulation:



2) Frictionless horizontal spring oscillation

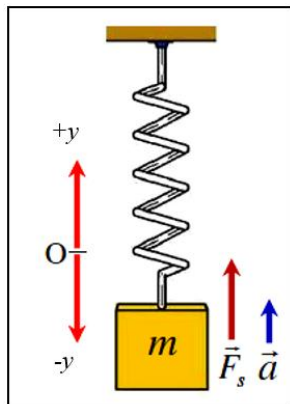


During the oscillation, are there any changes to the magnitude and direction of the acceleration? Explain.

Simulation:



3) Vertical spring oscillation



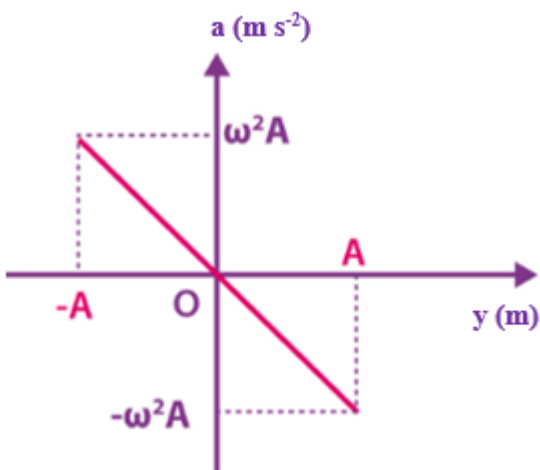
During the oscillation, are there any changes to the magnitude and direction of the acceleration? Explain.

Simulation:



Graph of acceleration against displacement, ($a - y$)

From the relationship between acceleration and displacement, $a = -\omega^2 y$, the graph of acceleration against displacement ($a - y$) is shown in figure below.

**Explanation:**

The straight line graph passing through the origin shows that **acceleration of a body is directly proportional to its displacement** from the equilibrium position (origin).

When the **displacement from equilibrium position is maximum (i.e. Amplitude)**, the **acceleration of motion is will be maximum**;
 $a_{max} = -\omega^2 A$.

The **negative slope** shows the **direction of a and y is opposite to each other**. From $a = -\omega^2 A$, the **gradient** of graph $a - y$ is always constant.

Example 1

A mass which hangs from the end of a vertical helical spring is in SHM of amplitude 2.0 cm. If three complete oscillations take 4.0 s, determine the magnitude of acceleration of the mass

- At the equilibrium position.
- When the displacement is maximum.

Answers:

a) $a = -\omega^2 y$

At equilibrium position, $y = 0$

$$a = 0 \text{ m s}^{-2}$$

b) three complete oscillations take 4.0 s $\rightarrow \omega = \frac{3 \text{ oscillations}}{4 \text{ s}} \times \frac{2\pi}{1}$
 $\omega = 1.5\pi \text{ rad s}^{-1}$

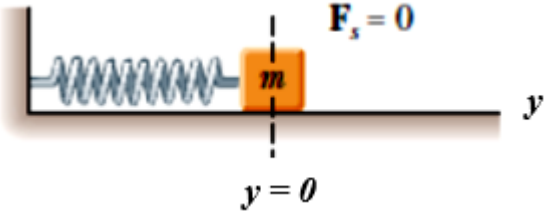
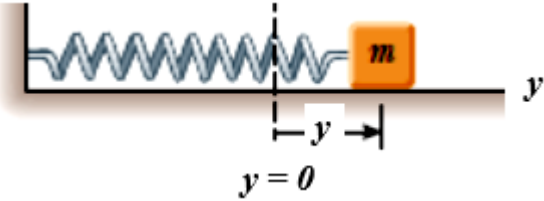
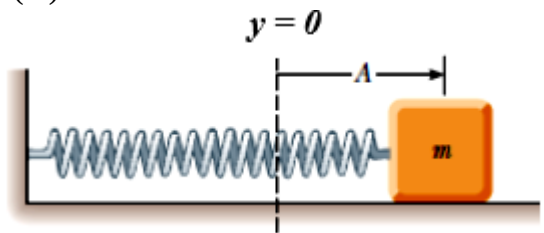
at maximum displacement, $y = A = 2.0 \text{ cm}$

$$a = -\omega^2 y$$

$$a = -(1.5\pi)^2 (2.0 \times 10^{-2})$$

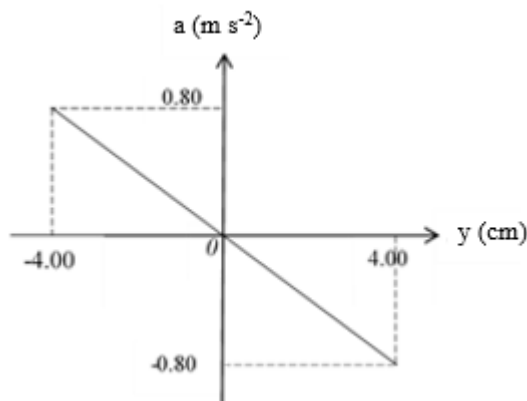
$$a = -0.44 \text{ m s}^{-2}$$

Terminology in Simple Harmonic Motion

Quantity	Definition
Equilibrium position	A point where the acceleration and force exerted on the body undergoing oscillation is zero ($y = 0$; $a = 0$; $F = 0$). 
Displacement, y	Distance from the equilibrium position. 
Amplitude, A	Is defined as the maximum magnitude of the displacement from the equilibrium position. SI unit is metre (m). 
Period, T	Is defined as the time taken for one complete cycle. SI Unit: second (s).
Frequency, f	Is defined as the number of complete cycles in one second. SI Unit: hertz (Hz) where $1 \text{ Hz} = 1 \text{ cycle s}^{-1} = 1 \text{ s}^{-1}$ $f = \frac{1}{T}$
Angular frequency, ω	Number of rotation in one second. SI Unit: radian per second (rad s^{-1}) $\omega = 2\pi f = \frac{2\pi}{T}$

Example 2

The figure shows the relationship between the acceleration and its displacement from a fixed point for a body of mass 2.50 kg at which executes SHM. Determine



- The amplitude of the motion.
- The maximum acceleration of the body in the motion.
- The angular frequency of the body in the motion.
- The period of the motion.

Answers:

a) From the graph, $A = 4.0 \text{ cm}$

d) $T = \frac{2\pi}{\omega}$
 $T = \frac{2\pi}{4.47}$
 $T = 1.41 \text{ s}$

b) From the graph, $a_{\max} = 0.8 \text{ m s}^{-2}$

c) $a = -\omega^2 y$
 $0.8 = -\omega^2 (-4.0 \times 10^{-2})$
 $\omega = 4.47 \text{ rad s}^{-1}$

Example 3

A particle in simple harmonic motion makes 20 complete oscillations in 5.0 s. Determine

- The angular frequency of the motion.
- The period of oscillations of the motion.

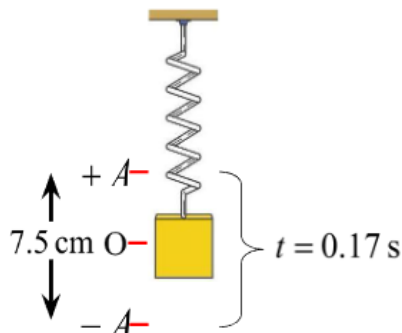
Answers:

a) 20 complete oscillations take 5.0 s $\rightarrow \omega = \frac{20 \text{ oscillations}}{5 \text{ s}} \times \frac{2\pi}{1}$
 $\omega = 8\pi \text{ rad s}^{-1}$

b) $T = \frac{2\pi}{\omega} = \frac{2\pi}{8\pi}$
 $T = 0.25 \text{ s}$

Example 4

A body hanging from one end of a vertical spring performs vertical SHM. The distance between two points, at which the acceleration of the body is maximum is 7.5 cm. If the time taken for the body to move between the two points is 0.17 s, determine



- The amplitude of the motion.
- The frequency of the motion.
- The maximum acceleration of body in the motion.

Answers:

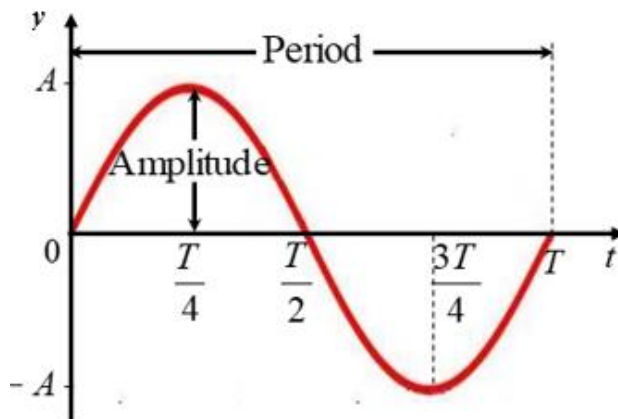
$$\begin{array}{lll}
 \text{a) } A = \frac{7.5 \text{ cm}}{2} = 3.75 \text{ cm} & \text{b) } T = 2(0.17 \text{ s}) & \text{c) } a_{\max} = \omega^2 A \\
 & T = 0.34 \text{ s} & a_{\max} = (2\pi f)^2 A \\
 & f = \frac{1}{T} & a_{\max} = [2\pi(2.94)]^2 (3.75 \times 10^{-2}) \\
 & f = \frac{1}{0.34} & a_{\max} = 12.8 \text{ m s}^{-2} \\
 & f = 2.94 \text{ Hz} &
 \end{array}$$

Graph of displacement against time, ($y - t$)

From the general equation of displacement as a function of time in SHM,

$$y = A \sin \omega t$$

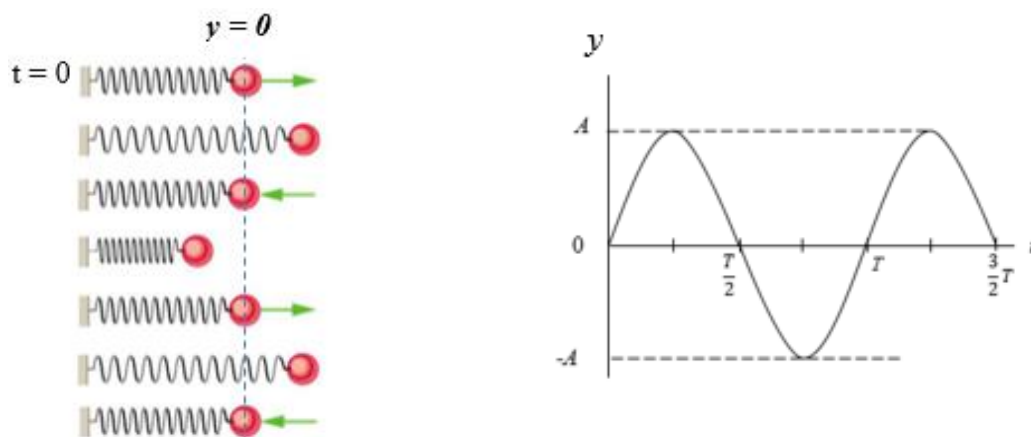
The displacement – time graph is shown in figure below:



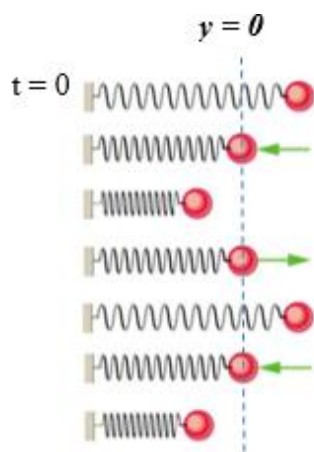
Question 1

Can you sketch the graph of displacement against time for each horizontal spring oscillation below? How do they differ?

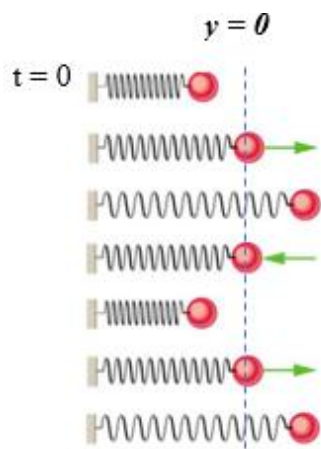
a)



b)

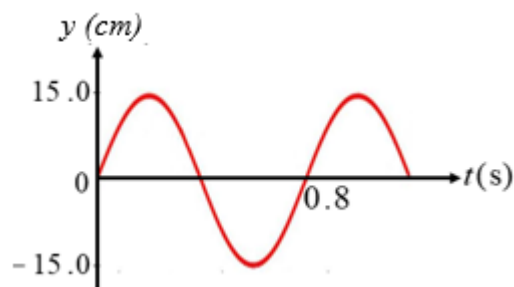


c)



Example 5

The displacement of an oscillating object as a function of time is shown in figure below.



From the graph, determine for these oscillations

- The amplitude, the period and the frequency.
- The angular frequency.
- The equation of displacement as a function of time.

Answers:

a) $A = 15.0 \text{ cm}$

$T = 0.8 \text{ s}$

$$f = \frac{1}{T} = \frac{1}{0.8} = 1.25 \text{ Hz}$$

b) $\omega = 2\pi f$

$\omega = 2\pi(1.25)$

$\omega = 2.5\pi \text{ rad s}^{-1}$

- c) By substituting $A = 15.0 \text{ cm}$ and $\omega = 2.5\pi \text{ rad s}^{-1}$ into the general equation,

$$y = 15 \sin 2.5\pi t$$

where y is in centimeters and t is in seconds

Example 6

A 0.2 kg mass oscillates according to the equation $y = 8 \sin 2.5\pi t$ where x is in centimeters and t is in seconds.

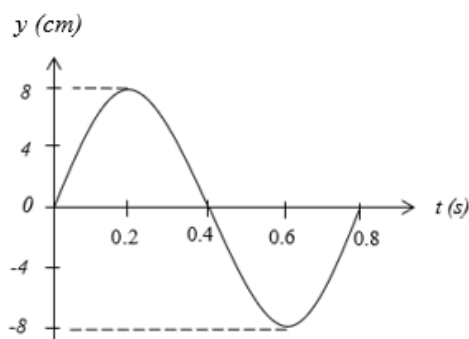
- Determine the amplitude and angular frequency of the oscillations.
- Sketch the displacement-time graph for one complete oscillation.

Answers:

- a) By comparing equation $y = 8 \sin 2.5\pi t$ with the general equation,

$A = 8.0 \text{ cm}$

$\omega = 2.5\pi \text{ rad s}^{-1}$



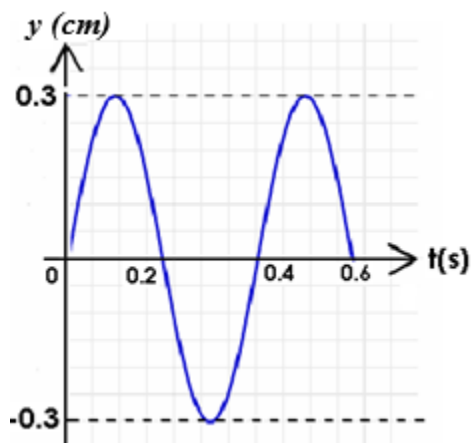
Velocity, v & Acceleration, a

Velocity, v	
Velocity, v as a function of time t From the definition of instantaneous velocity, $v = \frac{dy}{dt}$ $y = A \sin(\omega t)$ $v = \frac{d}{dt}(A \sin(\omega t))$ $v = A\omega \cos(\omega t) \quad \dots (1)$ Equation (1) is an equation of velocity as a function of time in SHM The maximum velocity, v_{max} occurs when $\cos(\omega t) = 1$, therefore, $v_{max} = A\omega$	Relationship between v and y From (1) $v = A\omega \cos(\omega t)$ And displacement is given by $y = A \sin(\omega t)$ $\sin(\omega t) = \frac{y}{A} \dots (2)$ From trigonometry identities $\sin^2 \theta + \cos^2 \theta = 1$ Therefore, $\cos \theta = \sqrt{1 - \sin^2 \theta}$ $\cos(\omega t) = \sqrt{1 - \sin^2(\omega t)} \dots (3)$ Substitute equation (3) and (2) into equation (1) $v = A\omega \sqrt{1 - \left(\frac{y}{A}\right)^2}$ $v = \omega \sqrt{A^2 - A^2 \left(\frac{y^2}{A^2}\right)}$ $v = \omega \sqrt{A^2 - y^2}$
Acceleration, a	
Acceleration as a function of time t From the definition of instantaneous acceleration. $a = \frac{dv}{dt}$ $v = A\omega \cos(\omega t)$ $a = \frac{d}{dt}(A\omega \cos(\omega t))$ $a = -A\omega^2 \sin(\omega t) \quad \dots (1)$ The maximum acceleration (most positive) $a_{max} = A\omega^2$	Relationship between a and y From equation (1) $a = -A\omega^2 \sin(\omega t)$ Displacement equation is given by $y = A \sin(\omega t) \dots (2)$ Substitute (2) into (1) $a = -\omega^2 y$

Example 7

Figure below shows how the displacement, y varies with time t for a system undergoing SHM. Determine

- The amplitude, the period and the frequency.
- The angular frequency.
- The displacement, velocity & acceleration for the system at time $t = 2$ s.
- The maximum speed.
- The maximum acceleration.



Answers:

a) $A = 0.3 \text{ cm}; T = 0.4 \text{ s}; f = \frac{1}{T} = \frac{1}{0.4} = 2.5 \text{ Hz}$

b) $\omega = 2\pi f = 2\pi(2.5) = 5\pi \text{ rad s}^{-1}$

c) From $y = A \sin \omega t$, we have $y = 0.3 \sin 5\pi t$

$$y = 0.3 \sin[5\pi(2)]$$

$$y = 0 \text{ cm}$$

From $v = A\omega \cos(\omega t)$, we have $v = (0.3)(5\pi) \cos(5\pi t)$

$$v = 1.5\pi \cos(5\pi t)$$

$$v = 1.5\pi \cos[5\pi(2)]$$

$$v = 1.5\pi \text{ cm s}^{-1}$$

From $a = -A\omega^2 \sin(\omega t)$, we have $a = -(0.3)(5\pi)^2 \sin(5\pi t)$

$$a = 7.5\pi^2 \sin(5\pi t)$$

$$a = 7.5\pi^2 \sin[5\pi(2)]$$

$$a = 0 \text{ cm s}^{-2}$$

*Make sure to set calculator in mode rad (radian)

d) $v_{\max} = A\omega = (0.3)(5\pi) = 1.5\pi \text{ cm s}^{-1}$

e) $a_{\max} = A\omega^2 = (0.3)(5\pi)^2 = 7.5\pi^2 \text{ cm s}^{-2}$

Chapter 1 MCQs Practice:



